

**EXAMINING LIFESTYLE FACTORS
AND METABOLIC HEALTH MARKERS
ASSOCIATED WITH DIABETES USING
NHANES DATA (2021–2023)**

A Research Project – I Report

Submitted in partial fulfilment of the requirements for the
Award of the degree of

Master of Computer Applications

By

Pargat Singh

322201210



**Centre for Distance and Online Education
LOVELY PROFESSIONAL UNIVERSITY
PHAGWARA, PUNJAB**

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DECLARATION BY THE STUDENT

To whom-so-ever it may concern

I, **Pargat Singh, 322201210**, hereby declare that the work done by me on **"Examining Lifestyle Factors and Metabolic Health Markers Associated with Diabetes Using NHANES Data (2021-2023)"**, is a record of original work for the partial fulfilment of the requirements for the award of the degree, Master of Computer Applications.

Pargat Singh (322201210)

Signature of the student:

Dated: 20/01/2026

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I would like to express my heartfelt gratitude to the faculty members of the MCA programme at Lovely Professional University for their continuous support and guidance throughout this research project. Their expertise and mentorship have been invaluable in shaping my understanding of data science and research methodology.

I am deeply grateful to the Centers for Disease Control and Prevention (CDC) for making the NHANES data freely available for research purposes. This comprehensive dataset has provided the foundation for exploring important questions about diabetes and lifestyle factors in the American population.

I would also like to thank my family and friends for their constant encouragement and understanding throughout this academic journey. Their support has been a source of strength and motivation.

ABSTRACT

Diabetes mellitus is one of the fastest-growing chronic diseases worldwide, affecting over 38 million Americans. Understanding the relationship between lifestyle factors and diabetes prevalence is essential for developing effective prevention strategies. This Research Project–I establishes the foundation for analyzing this relationship using current health surveillance data.

The study uses the National Health and Nutrition Examination Survey (NHANES) 2011–2018 dataset from the Centers for Disease Control and Prevention. This dataset provides comprehensive health information including demographic characteristics, metabolic health indicators such as BMI, fasting glucose, and HbA1c levels, and lifestyle factors including sedentary behavior, smoking status, and dietary intake.

This report accomplishes four main objectives. First, it reviews existing literature to define the research problem. Second, it identifies gaps in current knowledge about diabetes and lifestyle factors. Third, it formulates seven specific research questions addressing prevalence patterns, lifestyle associations, and health disparities. Fourth, it outlines the analytical methodology using R programming for statistical analysis.

The expected outcome is a well-defined research framework ready for implementation in Research Project–II. The findings will help identify which lifestyle and demographic factors show the strongest associations with diabetes, potentially informing prevention programs and health education strategies.

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LIST OF ABBREVIATIONS

BMI - Body Mass Index

CDC - Centers for Disease Control and Prevention

DXA - Dual-Energy X-ray Absorptiometry

HbA1c - Glycated Hemoglobin (Hemoglobin A1c)

NHANES - National Health and Nutrition Examination Survey

R - Statistical Programming Language

T2DM - Type 2 Diabetes Mellitus

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CHAPTER 1: INTRODUCTION

1.1 Background and Context

Diabetes is one of the most pressing health challenges facing the world today. It is a chronic metabolic disorder characterized by elevated blood glucose levels due to insufficient insulin production or improper insulin utilization by the body. This condition affects millions of people globally and creates serious health complications including cardiovascular disease, kidney failure, nerve damage, and vision loss.

In the United States, the diabetes epidemic has reached alarming proportions. More than 38 million Americans are currently living with diabetes, representing approximately 11.6 percent of the population². Perhaps even more concerning is that nearly one in five people with diabetes remain undiagnosed, missing crucial opportunities for early intervention and treatment². The economic burden is staggering, with diabetes-related healthcare costs exceeding \$327 billion annually.

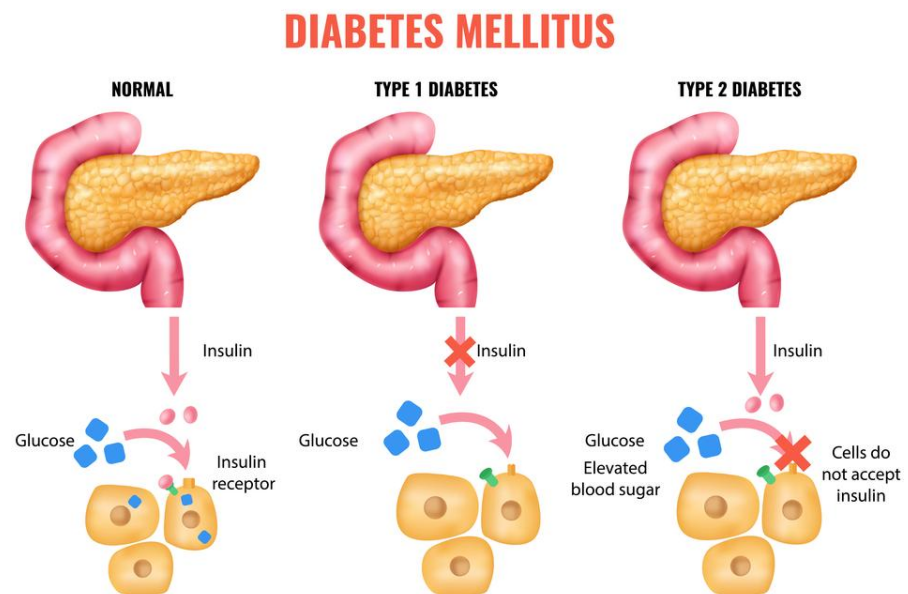


Figure 1.1: Mechanisms of Type 1 and Type 2 Diabetes⁶

Type 2 diabetes mellitus accounts for approximately 90 to 95 percent of all diabetes cases. Unlike Type 1 diabetes, which is typically diagnosed

in childhood and involves autoimmune destruction of insulin-producing cells, Type 2 diabetes develops gradually over time and is strongly influenced by lifestyle factors. This distinction is critical because it means Type 2 diabetes is largely preventable through modifications in diet, physical activity, and weight management.

Understanding how lifestyle choices affect diabetes risk is fundamental to prevention efforts. Factors such as sedentary behavior, dietary patterns, smoking status, and body weight all play important roles in determining who develops diabetes and who does not. However, the relative importance of these factors and their interactions in contemporary populations requires ongoing investigation using current data.

1.2 Importance of Data Science in Health Research

Modern healthcare generates vast amounts of data from surveys, electronic health records, laboratory tests, and epidemiological studies. However, raw data alone provides limited value. The real insights emerge through systematic analysis using statistical and computational methods.

Data science bridges the gap between data collection and actionable knowledge. It combines statistical analysis, programming skills, and domain expertise to extract meaningful patterns from complex datasets. For Master of Computer Applications students, applying these skills to real-world health problems demonstrates the practical value of computational methods beyond traditional software development.

This project utilizes R programming language, which has become the standard tool for statistical analysis in research and industry. R provides comprehensive libraries for data manipulation, statistical testing, and visualization⁵. Learning to work with real epidemiological data develops practical skills that are highly valued in healthcare analytics, public health institutions, pharmaceutical companies, and health technology organizations.

1.3 About the NHANES Dataset

The National Health and Nutrition Examination Survey (NHANES) is a cornerstone of public health surveillance in the United States. Conducted by the Centers for Disease Control and Prevention, NHANES has been collecting health information about Americans since the 1960s. The survey became a continuous program in 1999, releasing new data in two-year cycles.

What distinguishes NHANES from simple questionnaire surveys is its comprehensive approach. Participants undergo complete physical examinations, laboratory testing, and detailed interviews. This multi-modal data collection provides objective measurements alongside self-reported information. The survey captures demographics, medical history, dietary intake, physical activity patterns, body measurements, blood tests, and numerous other health indicators.

For this project, I am analyzing data from the 2021-2023 cycle, which represents the most recent complete dataset available². This timeframe is particularly relevant as it captures health behaviors and outcomes during the post-COVID-19 pandemic period, when many Americans experienced significant lifestyle changes. The dataset includes information from thousands of participants selected through a complex probability sampling design to ensure national representativeness.

1.4 Significance of This Research

This research addresses several important needs in diabetes epidemiology and prevention science. First, diabetes prevalence continues to rise despite decades of public health efforts¹. Understanding current patterns and risk factors is essential for developing more effective interventions. Using the most recent data available ensures our findings reflect contemporary health behaviors rather than outdated patterns.

Second, lifestyle habits have evolved significantly in recent years. The COVID-19 pandemic accelerated remote work adoption, potentially

increasing sedentary behavior. Food delivery services have changed dietary patterns. These shifts necessitate updated analyses of how contemporary lifestyle factors relate to diabetes risk.

Third, this project demonstrates technical competence in data science methodologies. It showcases abilities in data acquisition, data cleaning, statistical analysis, hypothesis testing, and result interpretation. These are core competencies for any data-focused career in healthcare or technology sectors.

Finally, the research addresses health equity concerns. Diabetes disproportionately affects certain racial and ethnic minority populations². By examining these disparities in current data, we can identify groups that would benefit most from targeted prevention programs and better understand potential barriers to diabetes prevention in vulnerable communities.

1.5 Research Questions

This study is guided by seven specific research questions that systematically examine different aspects of diabetes prevalence and lifestyle factors:

- What is the prevalence of diabetes in the NHANES 2021-2023 sample, and how does it vary by age, gender, and race/ethnicity?
- Is there a significant relationship between sedentary behavior (minutes per day) and diabetes status?
- Does current smoking status relate to diabetes prevalence?
- Is daily carbohydrate intake significantly different between individuals with and without diabetes?
- How do metabolic markers (BMI, fasting glucose, HbA1c) compare between diabetes and non-diabetes groups?
- Are there significant racial and ethnic disparities in diabetes prevalence and metabolic health indicators?

- Which lifestyle and demographic factors show the strongest statistical associations with diabetes?

CHAPTER 2: REVIEW OF LITERATURE

2.1 Introduction

Before undertaking original analysis, understanding the existing body of research is essential. This chapter reviews relevant literature on diabetes prevalence, lifestyle risk factors, and analytical approaches to NHANES data. The review establishes what is currently known, identifies gaps in knowledge, and provides context for the current study.

2.2 Diabetes Prevalence Trends

Long-term analysis of NHANES data reveals concerning trends in diabetes prevalence. Wang and colleagues examined data spanning from 1999 to 2018 and documented a steady increase in diagnosed diabetes among U.S. adults¹. Their study showed prevalence rising from about 10.3 percent in 1999-2000 to over 14.3 percent in 2017-2018. This upward trajectory persisted across most demographic subgroups, suggesting systemic factors driving the diabetes epidemic.

Even more concerning, the researchers found that only about 21 percent of people with diagnosed diabetes were meeting all their health goals¹. These goals include keeping blood sugar, blood pressure, and cholesterol at healthy levels. This indicates that diabetes is not only becoming more common, but many people are also struggling to manage the condition effectively.

The most recent surveillance data from the National Center for Health Statistics, covering August 2021 through August 2023, provides updated prevalence estimates². During this period, approximately 15.9 percent of U.S. adults had diabetes, with 10.8 percent having been diagnosed and an additional 5.1 percent remaining undiagnosed. The persistence of substantial undiagnosed diabetes underscores the importance of screening programs and public awareness campaigns.

2.3 Physical Activity and Sedentary Behavior

The relationship between physical activity and diabetes risk has been extensively studied. A prospective cohort study from the Multi-Ethnic Study of Atherosclerosis followed nearly 6,000 adults for over 11 years to examine how walking pace and volume related to diabetes incidence³. The researchers found that people who walked at a brisk pace had lower diabetes risk compared to those who walked slowly or did not walk at all, even after adjusting for total walking time and other confounding factors.

Importantly, this study also examined sedentary behavior measured through accelerometry. Participants who spent more time in sedentary activities, particularly television viewing, had markedly higher diabetes risk compared to those with less sedentary time³. This association remained significant even when accounting for moderate-to-vigorous physical activity levels, suggesting that prolonged sitting independently contributes to diabetes risk.

Hamilton and colleagues reviewed the physiological mechanisms underlying sedentary behavior's harmful effects⁴. When large skeletal muscle groups remain inactive for extended periods, metabolic processes that regulate blood glucose become impaired. Muscle contraction stimulates glucose uptake independent of insulin, so prolonged sitting reduces this important regulatory pathway. Additionally, sedentary time is associated with reduced lipoprotein lipase activity, contributing to metabolic dysfunction.

2.4 Obesity and Body Mass Index

The connection between excess body weight and diabetes is among the most robust findings in metabolic epidemiology. Recent NHANES data demonstrates a strong dose-response relationship, with diabetes prevalence increasing progressively across weight categories². Among adults with normal body weight, approximately 7.4 percent have diabetes. This rises to 12.2 percent among overweight individuals and reaches 24.2 percent among those with obesity.

The biological mechanisms linking obesity to diabetes involve multiple pathways. Adipose tissue, particularly visceral fat accumulated around internal organs, functions as an endocrine organ secreting inflammatory cytokines and hormones that interfere with insulin signaling. This insulin resistance forces the pancreas to produce increasing amounts of insulin to maintain normal blood glucose levels. Eventually, pancreatic beta cells become exhausted and cannot meet the elevated insulin demand, resulting in hyperglycemia and diabetes.

Body Mass Index, calculated as weight in kilograms divided by height in meters squared, serves as a practical screening tool for population studies. While BMI cannot distinguish between muscle mass and fat mass or identify fat distribution patterns, it correlates reasonably well with body fatness at the population level and predicts diabetes risk effectively in epidemiological research.

2.5 Racial and Ethnic Disparities

Diabetes prevalence varies substantially across racial and ethnic groups in the United States. The most recent national data shows that American Indian and Alaska Native adults experience the highest diabetes burden at approximately 14.5 percent, followed by Non-Hispanic Black adults and Hispanic adults both around 12.1 percent. In contrast, Non-Hispanic White adults have a prevalence of approximately 7.4 percent².

These disparities persist even after accounting for socioeconomic factors such as education and income, suggesting complex interplay of social, environmental, and possibly genetic factors. Communities of color often face structural barriers to healthy living including limited access to affordable nutritious food, fewer safe recreational spaces for physical activity, and reduced access to quality healthcare services.

Chronic stress related to experiences of discrimination and socioeconomic disadvantage may also contribute to metabolic dysregulation through neuroendocrine pathways. Cultural factors including traditional dietary practices and attitudes toward physical

activity add additional complexity to understanding and addressing these disparities.

2.6 Analytical Approaches to NHANES Data

Working with NHANES data requires understanding of survey methodology and appropriate analytical techniques. NHANES employs a complex, stratified, multistage probability sampling design to ensure results represent the non-institutionalized U.S. civilian population. This sampling approach means that standard statistical methods assuming simple random sampling may produce incorrect results if applied without modification.

For this project, I take a simplified analytical approach appropriate for an educational research project. Rather than incorporating the complex survey weights and design variables, I treat the sample as representative of the population and apply standard statistical methods⁵. This approach is acceptable for learning purposes and for identifying associations, though formal population estimates would require proper survey weighting procedures.

The R programming environment offers excellent tools for NHANES analysis. Packages such as `nhanesA` facilitate data downloading, while `dplyr` and `tidyr` enable efficient data manipulation⁵. For statistical testing, packages like `rstatix` provide user-friendly functions for common tests including Wilcoxon rank-sum tests, chi-square tests, and Kruskal-Wallis tests with appropriate post-hoc comparisons.

2.7 Gaps in Current Knowledge

Despite extensive research on diabetes epidemiology, several gaps motivate the current study. First, most published analyses use data from earlier survey cycles. The 2021-2023 cycle is the most recent complete dataset, and patterns may have evolved, particularly following the COVID-19 pandemic which significantly disrupted lifestyle behaviors for many Americans.

Second, many studies examine individual risk factors in isolation. Analyzing multiple factors simultaneously, including sedentary behavior, smoking, diet, BMI, and demographic characteristics, provides a more comprehensive picture of diabetes epidemiology. Understanding which factors show the strongest associations when considered together helps prioritize prevention efforts.

Third, while racial disparities are well-documented, continuous monitoring is necessary to assess whether public health interventions are reducing these gaps or if disparities persist or widen. The current analysis examines these patterns in the most recent available data.

Finally, from an educational perspective, there is value in replicating and extending published analyses using current data. This develops practical research skills, reinforces statistical concepts, and contributes to ongoing surveillance of this critical public health issue.

CHAPTER 3: PROBLEM FORMULATION AND RESEARCH TITLE

3.1 Justification for the Study

The need for this study emerges from several converging factors. Diabetes continues to pose an enormous public health burden with prevalence rates climbing steadily over the past two decades¹. The economic costs exceed hundreds of billions of dollars annually in the United States alone, while the human costs in terms of reduced quality of life and premature mortality are immeasurable.

Recent years have witnessed significant shifts in American lifestyles. The COVID-19 pandemic accelerated trends toward remote work and reduced physical activity. Food delivery services have proliferated, potentially affecting dietary patterns. These changes make it essential to examine diabetes risk factors using the most current data available rather than relying solely on pre-pandemic studies.

From an educational standpoint, this project demonstrates mastery of data science techniques applied to real-world health data. It showcases competence in data acquisition, preparation, statistical analysis, and interpretation. These are foundational skills for careers in health informatics, biostatistics, epidemiology, and data-driven healthcare innovation.

Additionally, examining health disparities addresses critical equity concerns². If certain populations bear disproportionate diabetes burden, identifying these patterns can inform targeted interventions and help address social determinants of health that contribute to inequalities.

3.2 Problem Statement

The central problem this research addresses is understanding the contemporary relationship between modifiable lifestyle factors and diabetes prevalence using the most recent nationally representative health surveillance data. While diabetes epidemiology has been

extensively studied, several critical gaps motivate this investigation.

First, we must establish current baseline estimates for diabetes prevalence across demographic subgroups using post-pandemic data. Second, quantifying associations between specific behavioral factors—such as daily sedentary time, smoking habits, and dietary patterns—and diabetes status requires examination in current populations. Third, comparing objective metabolic health indicators between individuals with and without diabetes provides insight into physiological differences that may inform screening protocols. Finally, identifying persistent health disparities across racial and ethnic groups remains essential for developing equitable prevention strategies.

This study addresses these needs through systematic analysis answering the seven research questions outlined in Chapter 1, Section 1.5. Rather than testing predetermined hypotheses, this exploratory investigation examines patterns and associations in observed data, establishing which factors demonstrate statistical relationships with diabetes prevalence worthy of deeper investigation in future research.

3.3 Research Title

Based on the problem statement and research objectives, the title for this project is:

Examining Lifestyle Factors and Metabolic Health Markers Associated with Diabetes Using NHANES Data (2021-2023)

This title effectively communicates several key aspects of the research. The word "Examining" indicates that this is an observational study examining associations rather than establishing causation. "Lifestyle Factors" encompasses the multiple behavioral and health-related variables being investigated, including sedentary behavior, smoking status, and dietary intake. "Metabolic Health Markers" refers to the physiological indicators such as BMI, fasting glucose, and HbA1c levels that will be analyzed. "Associated with Diabetes" clarifies that the

outcome of interest is the presence or absence of diabetes in the population. Finally, specifying "NHANES Data (2021-2023)" identifies both the data source and the specific time period, emphasizing the currency of the analysis and its relevance to contemporary health patterns.

CHAPTER 4: AIMS AND OBJECTIVES

4.1 Overall Aim

The overarching aim of this research is to systematically analyze the relationship between lifestyle factors and diabetes prevalence in the contemporary United States population using the NHANES 2021-2023 dataset. This involves describing current diabetes patterns, testing statistical associations between lifestyle variables and diabetes status, and examining health disparities across demographic groups.

4.2 Specific Objectives

To achieve the overall aim, this study has four specific objectives:

Objective 1: Data Acquisition and Preparation

The first objective is to obtain and prepare a clean, analysis-ready dataset from NHANES 2021-2023⁵. This involves downloading the appropriate data files, merging information from multiple sources, selecting relevant variables, handling missing data appropriately, and creating derived variables as needed. The final dataset must include demographics (age, gender, race/ethnicity), diabetes status, metabolic markers (BMI, fasting glucose, HbA1c), and lifestyle factors (sedentary activity minutes, smoking status, carbohydrate intake).

Objective 2: Descriptive Analysis of Diabetes Patterns

The second objective is to describe diabetes prevalence and related characteristics in the sample. This includes calculating overall diabetes prevalence, stratifying prevalence by age groups, gender, and race/ethnicity, computing summary statistics (means and standard deviations) for continuous variables in diabetes and non-diabetes groups, and creating preliminary visualizations to identify patterns in the data.

Objective 3: Statistical Testing of Associations

The third objective is to formally test whether lifestyle factors and demographic characteristics are significantly associated with diabetes status. For continuous variables, this involves conducting Wilcoxon rank-sum tests to compare distributions between diabetes and non-diabetes groups. For categorical variables, chi-square tests of independence will assess associations. All tests will use appropriate adjustments for multiple comparisons to control the false discovery rate.

Objective 4: Analysis of Racial and Ethnic Disparities

The fourth objective specifically examines health disparities across racial and ethnic groups². This involves testing whether diabetes prevalence differs significantly by race/ethnicity, comparing metabolic markers (HbA1c, fasting glucose, BMI) across racial groups using Kruskal-Wallis tests, conducting post-hoc pairwise comparisons to identify which specific groups differ, and interpreting findings in the context of health equity.

4.3 Expected Outcomes

Upon completion of these objectives, this study will produce several concrete outcomes. These include a clean, well-documented dataset ready for analysis; comprehensive descriptive statistics characterizing the sample; statistical evidence regarding which lifestyle factors show significant associations with diabetes; identification of groups experiencing health disparities; and professional visualizations communicating key findings. These outputs will form the foundation for Research Project II, where detailed results will be presented and discussed in depth.

CHAPTER 5: IMPLEMENTATION OF RESEARCH PROJECT

5.1 Overview of Implementation

This chapter describes the systematic implementation of the research project, covering all steps from data acquisition through statistical analysis. The implementation follows a structured workflow: first, downloading and importing multiple NHANES data files; second, merging these files into a single dataset; third, cleaning and preparing the data for analysis; fourth, conducting descriptive statistical analyses; and finally, performing hypothesis testing and specialized analyses.

All data processing and statistical analysis procedures described in this chapter were implemented using R version 4.5.1 running on Windows x64 platform. The complete, well-documented R script is publicly available for reproducibility and verification purposes at:

<https://github.com/pargat-singhh/nhanes-diabetes-analysis>

The script includes detailed comments explaining each step of the analysis workflow, from data acquisition through final visualization generation.

5.2 Required R Packages

The analysis utilized several R packages, each serving specific purposes:

- `haven`: Reading SAS transport files (.xpt format) from NHANES
- `dplyr`: Data manipulation including filtering, selecting, and joining datasets
- `tidyr`: Data reshaping and handling missing values
- `rstatix`: Statistical testing with clean output formatting
- `ggplot2`, `ggpubr`, `ggcorrplot`: Data visualization

5.3 Data Acquisition

NHANES data are publicly available from the CDC website. For the 2021-2023 cycle, data are organized into multiple files covering different domains. I downloaded the following data files:

Table 5.1 presents NHANES 2021–2023 Data Files Used in the Study:

File Name	NHANES Component	Description / Variables Used
DEMO_L.XPT	Demographics	Age (years), Gender, Race/Ethnicity
DIQ_L.XPT	Diabetes Questionnaire	Self-reported diabetes status
GLU_L.XPT	Laboratory Data	Fasting plasma glucose (mg/dL)
GHB_L.XPT	Laboratory Data	Glycohemoglobin (HbA1c, %)
PAQ_L.XPT	Physical Activity Questionnaire	Daily sedentary behavior (minutes/day)
SMQ_L.XPT	Smoking Questionnaire	Current smoking status
BMX_L.XPT	Body Measurements	Body Mass Index (BMI, kg/m ²)
DR1TOT_L.XPT	Dietary Interview (Day 1)	Total carbohydrate intake (grams/day)

Each file was imported using the `read_xpt` function from the `haven` package, which correctly handles the SAS transport format used by NHANES.

5.4 Data Merging

All NHANES files contain a unique participant identifier called `SEQN` (sequence number). I used this variable to merge data from different files using left joins, starting with demographics as the base dataset. This

approach ensures that all participants in the demographic file are retained, with their corresponding data from other files added as available.

During the merging process, I selected only the relevant variables from each file and renamed them to more intuitive names. For example, RIDAGEYR became Age, BMXBMI became BMI, and PAD680 became Sedentary_activity_mins.

5.5 Data Cleaning

Data cleaning involved several critical steps to ensure analysis quality. First, I removed participants with missing values in any of the key variables required for analysis. This complete-case analysis approach, while potentially reducing sample size, ensures that all participants included have complete information for all variables of interest.

Second, I filtered out invalid or missing codes. In NHANES, the value 9999 in sedentary activity minutes indicates "Don't know" responses, which were excluded. For diabetes status, I removed participants coded as 3 (borderline diabetes) to create clear diabetes versus non-diabetes groups.

Third, I created new variables as needed. The diabetes variable was recoded from numeric values (1=Yes, 2=No) to meaningful factor labels ("Diabetes", "No_Diabetes"). Smoking status was recoded from numeric codes to descriptive categories ("Every Day", "Some Days", "Not at All"). This recoding makes results more interpretable.

After all cleaning steps, the final analysis dataset contained 963 participants with complete data on all variables. This represents approximately 10-15 percent of the full NHANES 2021-2023 sample, which is typical when requiring complete data across multiple domains including laboratory tests, questionnaires, and physical measurements.

5.6 Descriptive Statistics

Descriptive analysis began by calculating diabetes prevalence overall and by demographic subgroups. For continuous variables, I computed means and standard deviations separately for participants with and without diabetes. This provides initial insight into how these variables differ between groups before formal statistical testing.

The dplyr package's `group_by` and `summarise` functions facilitated these calculations efficiently. By grouping the data by diabetes status and then applying summary functions, I generated comprehensive descriptive statistics tables showing the characteristics of each group.

5.7 Statistical Testing Methods

For comparing continuous variables between diabetes and non-diabetes groups, I used the Wilcoxon rank-sum test (also known as the Mann-Whitney U test). This non-parametric test was chosen because many health variables show skewed distributions that violate the normality assumption required for t-tests. The Wilcoxon test makes no distribution assumptions and is robust to outliers.

For categorical variables including gender, smoking status, and race/ethnicity, I used chi-square tests of independence to assess whether distributions differ by diabetes status. These tests evaluate whether the observed frequencies in contingency tables deviate significantly from what would be expected if the variables were independent.

To examine racial disparities in metabolic markers, I employed Kruskal-Wallis tests, which extend the Wilcoxon test to more than two groups. When these tests showed significant overall differences, I conducted Dunn's post-hoc tests with Benjamini-Hochberg adjustment to identify which specific racial groups differed from each other while controlling the false discovery rate.

Multiple comparison correction is crucial when conducting numerous statistical tests simultaneously. Without adjustment, the probability of at least one false positive result increases with the number of tests. I

applied Benjamini-Hochberg correction to control the false discovery rate at 5 percent, providing a balance between Type I and Type II error control.

5.8 Visualization Methods

Data visualization serves multiple purposes: exploring patterns, identifying outliers, and communicating results. I created several types of visualizations using ggplot2 and related packages:

- Box plots with jittered points showing distributions of continuous variables by diabetes status
- Correlation heatmaps displaying relationships among continuous variables
- Box plots by race/ethnicity showing disparities in metabolic markers

Box plots effectively display the distribution including median, quartiles, and outliers. Adding jittered points overlays individual data points, revealing sample sizes and data density. This combination provides both summary information and granular detail.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Sample Characteristics

The final analysis dataset comprised 963 participants with complete data on all variables of interest. Of these, 783 (81.3%) did not have diabetes while 180 (18.7%) had diabetes. This prevalence is consistent with national estimates from the same period, suggesting that despite requiring complete data across multiple domains, the sample remains reasonably representative.

6.2 Descriptive Statistics by Diabetes Status

Table 6.1 presents mean values and standard deviations for continuous variables stratified by diabetes status:

Variable	No Diabetes (n=783)	Diabetes (n=180)
Age (years)	56.9 ± 15.8	63.8 ± 10.9
BMI (kg/m ²)	29.0 ± 6.2	32.4 ± 8.0
Fasting Glucose (mg/dL)	103 ± 18.6	159 ± 64.1
HbA1c (%)	5.53 ± 0.58	7.40 ± 1.69
Sedentary Time (min/day)	349 ± 256	364 ± 245
Carbohydrate Intake (g/day)	253 ± 123	241 ± 115

Several notable patterns emerge from these descriptive statistics. Participants with diabetes were on average nearly 7 years older than those without diabetes (63.8 vs. 56.9 years). They had higher BMI (32.4 vs. 29.0 kg/m²), substantially elevated fasting glucose (159 vs. 103 mg/dL), and markedly higher HbA1c levels (7.4 vs. 5.5%). Interestingly, sedentary activity minutes were similar between groups (349 vs. 364 minutes), and carbohydrate intake was actually slightly lower in the diabetes group (241 vs. 253 grams).

6.3 Statistical Test Results

Wilcoxon rank-sum tests revealed highly significant differences for most variables. Age showed a very strong association with diabetes ($p < 0.001$), as did BMI ($p < 0.001$). The strongest associations were observed for the metabolic markers: fasting glucose ($p < 0.001$) and HbA1c ($p < 0.001$), which is expected since these directly measure glycemic control.

Surprisingly, sedentary activity minutes showed no significant difference between groups ($p = 0.517$). This contradicts some previous literature suggesting that sedentary behavior is associated with diabetes risk. However, this null finding might reflect limitations of self-reported sedentary time, or it could indicate that in the presence of other strong risk factors like obesity, sedentary behavior's independent effect becomes less apparent.

Carbohydrate intake showed a statistically significant difference ($p = 0.001$), but the effect was in the opposite direction from what might be expected: people with diabetes actually consumed less carbohydrate on average. This likely reflects dietary modifications made after diabetes diagnosis rather than indicating that lower carbohydrate intake increases diabetes risk.

Chi-square tests for categorical variables revealed that gender was significantly associated with diabetes ($p = 0.007$), with males having higher prevalence. However, smoking status showed no significant association ($p = 0.884$), suggesting that in this sample, smoking does not differentiate between diabetes and non-diabetes groups. Race/ethnicity showed a significant association ($p < 0.001$), consistent with known health disparities.

6.4 Metabolic Indicators Visualizations

Figure 6.1 displays distributions of three key metabolic indicators by diabetes status:

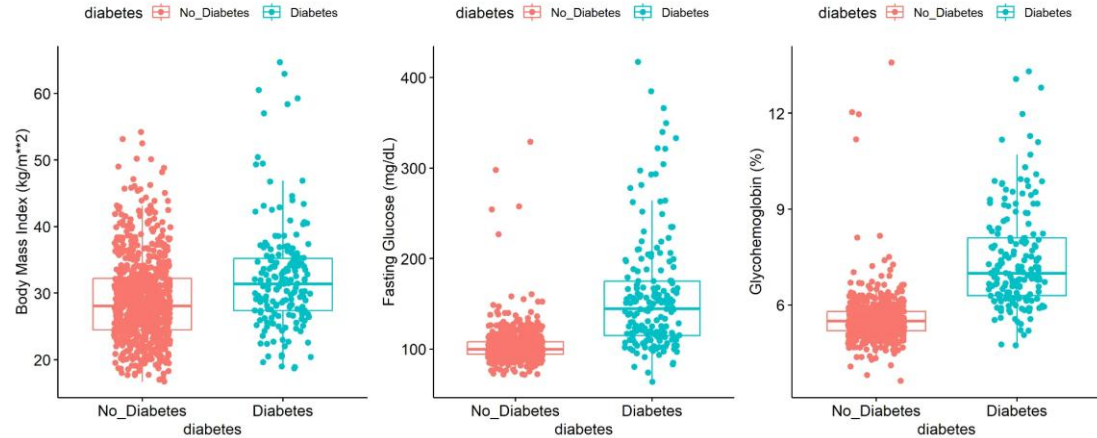


Figure 6.1: BMI, Fasting Glucose, and HbA1c by Diabetes Status

The box plots clearly illustrate the separation between groups. For BMI, while there is substantial overlap, the diabetes group shows a higher median and more outliers at the upper end. The glucose and HbA1c plots show even more dramatic separation, with the diabetes group having distinctly elevated values as expected given that these are diagnostic criteria for diabetes.

Figure 6.2 shows age and sedentary activity distributions:

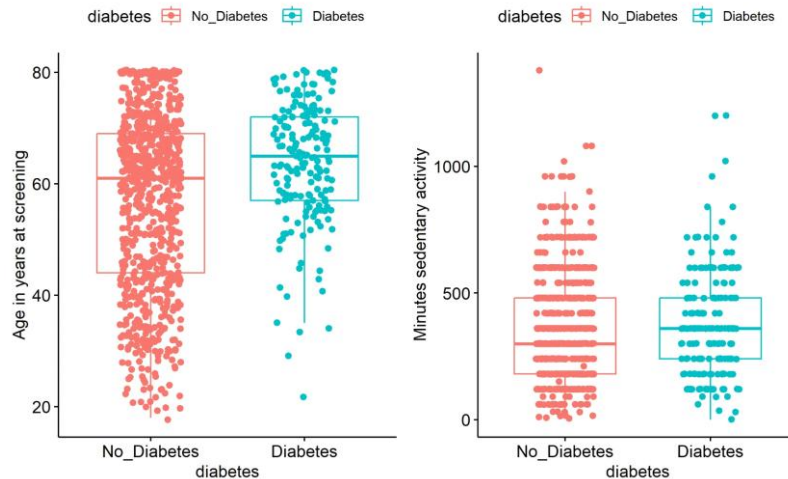


Figure 6.2: Age and Sedentary Activity by Diabetes Status

The age distribution shows that participants with diabetes tend to be older, with the diabetes group median around age 65 compared to approximately 60 for the non-diabetes group. The sedentary activity plot

shows considerable overlap between groups, visually confirming the statistical finding of no significant difference.

6.5 Correlation Analysis

Figure 6.3 presents a correlation heatmap showing relationships among continuous variables:



Figure 6.3: Correlation Matrix of Continuous Variables

The strongest correlation in the dataset is between fasting glucose and HbA1c ($r = 0.85$), which is expected since both measure glycemic status, though over different time scales. BMI shows modest positive correlations with age (0.13), fasting glucose (0.18), and HbA1c (0.16), consistent with obesity being a diabetes risk factor. Notably, sedentary activity minutes shows virtually no correlation with any other variable, including age ($r = -0.04$) and carbohydrate intake ($r = 0.04$). This may reflect the challenge of accurately measuring sedentary behavior through self-report.

6.6 Racial and Ethnic Disparities

Kruskal-Wallis tests revealed significant racial differences in HbA1c levels ($p < 0.001$) but not in fasting glucose ($p = 0.121$) or BMI ($p = 0.126$). This suggests that racial disparities in diabetes may be reflected

more consistently in long-term glycemic control (HbA1c) than in single fasting glucose measurements.

Figure 6.4 visualizes HbA1c distributions across racial groups:

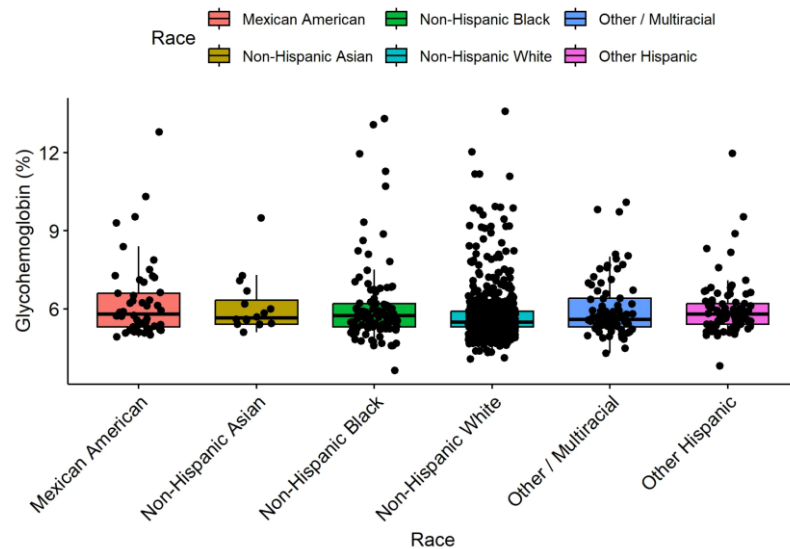


Figure 6.4: HbA1c Distribution by Race/Ethnicity

Post-hoc Dunn's tests identified significant pairwise differences. Mexican Americans had significantly higher HbA1c compared to Non-Hispanic Whites ($p = 0.049$), and Non-Hispanic Whites had significantly lower HbA1c compared to Other Hispanic participants ($p = 0.049$). These findings are consistent with documented disparities showing higher diabetes burden among Hispanic populations.

Interestingly, while Non-Hispanic Black participants are known to have higher diabetes prevalence nationally, in this sample their HbA1c levels did not differ significantly from other groups after multiple comparison adjustment. This might reflect sample size limitations or differences between diagnosed diabetes prevalence and glycemic control among those with diabetes.

6.7 Discussion of Key Findings

This analysis of NHANES 2021-2023 data confirms several well-established associations while revealing some unexpected patterns. The

strong relationships between diabetes and age, BMI, and metabolic markers align with extensive previous research. Older age and higher body weight remain dominant risk factors for Type 2 diabetes in the contemporary U.S. population.

The null finding for sedentary behavior contrasts with some literature but may reflect measurement challenges. Self-reported sedentary time is subject to recall bias and may not capture the nuances of sedentary patterns that objectively measured accelerometry reveals. Alternatively, the cross-sectional nature of this analysis means we cannot determine whether sedentary behavior preceded diabetes or vice versa.

The finding that people with diabetes consume less carbohydrate illustrates a common issue in cross-sectional research: reverse causation. People diagnosed with diabetes receive dietary counseling encouraging reduced carbohydrate intake. Thus, the observed lower carbohydrate consumption likely represents treatment adherence rather than a causal factor. This underscores the limitation of cross-sectional designs in inferring causation.

The persistence of racial disparities in HbA1c levels despite decades of public health efforts is concerning. These disparities likely reflect complex interactions among genetic factors, social determinants of health, healthcare access, and systemic barriers to healthy lifestyle maintenance. Addressing these inequities requires comprehensive approaches targeting both individual-level factors and structural determinants.

6.8 Limitations

Several limitations must be acknowledged. First, the cross-sectional design precludes causal inference. We can identify associations but cannot determine whether lifestyle factors preceded diabetes development or were modified in response to diagnosis. Longitudinal designs are needed to establish temporal relationships.

Second, requiring complete data across multiple domains substantially reduced the sample size. The 963 participants included represent only a fraction of the full NHANES sample. While this approach ensures data quality, it may introduce selection bias if participants with complete data differ systematically from those excluded.

Third, this analysis did not incorporate NHANES survey weights or complex sampling design. For educational purposes, treating the sample as a simple random sample is acceptable, but population-level estimates would require proper weighting procedures to account for unequal sampling probabilities and non-response.

Fourth, self-reported measures including sedentary behavior and dietary intake are subject to recall bias and social desirability effects. Participants may underreport sedentary time or overreport healthy behaviors, potentially weakening observed associations.

Finally, this analysis focused on diabetes presence/absence without distinguishing Type 1 from Type 2 diabetes or examining diabetes duration and severity. More nuanced analyses could provide additional insights into diabetes epidemiology.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1 Summary of Findings

This research successfully examined relationships between lifestyle factors and diabetes prevalence using the most recent NHANES data (2021-2023). The analysis of 963 participants with complete data revealed several important patterns:

- Diabetes prevalence in the analysis sample was 18.7%, consistent with national estimates.
- Age, BMI, fasting glucose, and HbA1c showed highly significant associations with diabetes status.
- Sedentary behavior did not differ significantly between diabetes and non-diabetes groups.
- Smoking status showed no significant association with diabetes in this sample.
- Carbohydrate intake was lower in the diabetes group, likely reflecting dietary modifications after diagnosis.
- Significant racial disparities were observed in HbA1c levels, with Mexican Americans and Other Hispanic participants showing higher values compared to Non-Hispanic Whites.

7.2 Implications for Public Health

These findings have several practical implications. First, they confirm that age and obesity remain the dominant modifiable risk factors for diabetes in the contemporary U.S. population. Prevention efforts should continue emphasizing weight management through dietary modification and physical activity promotion, particularly as individuals enter middle age.

Second, the persistence of racial disparities in glycemic control indicates ongoing need for culturally tailored interventions and efforts to address social determinants of health. Healthcare systems should prioritize

reducing barriers to diabetes management in communities experiencing higher disease burden.

Third, the lack of sedentary behavior association in this sample suggests that simply measuring total sedentary time may be insufficient. Future research and interventions might benefit from focusing on specific contexts of sedentary behavior (workplace versus leisure) or patterns of interrupting prolonged sitting rather than total minutes.

7.3 Implications for Data Science Education

From an educational perspective, this project demonstrated practical application of data science skills to meaningful health questions. Key competencies developed include data acquisition from public databases, data cleaning and preparation, statistical analysis using appropriate methods, data visualization for exploration and communication, and interpretation of results in context.

The project showcased the importance of domain knowledge in data analysis. Understanding diabetes epidemiology guided variable selection, informed choice of statistical methods, and enabled appropriate interpretation of unexpected findings like the reverse association with carbohydrate intake. Data science is most powerful when technical skills are combined with subject matter expertise.

7.4 Future Research Directions

This work suggests several promising directions for future investigation. First, longitudinal analysis following participants over time could clarify causal relationships between lifestyle factors and diabetes incidence. NHANES has followed some participants for multiple cycles, enabling such analyses.

Second, machine learning approaches could build predictive models identifying individuals at high diabetes risk based on easily measurable characteristics. Such models could inform targeted screening and

prevention programs, though careful validation would be essential to ensure performance across diverse populations.

Third, deeper investigation of racial disparities could examine potential mediating factors including healthcare access, health insurance coverage, neighborhood characteristics, and experiences of discrimination. Understanding mechanisms underlying disparities is essential for developing effective interventions.

Fourth, incorporating objectively measured physical activity and sedentary behavior from accelerometry data could provide more accurate assessments than self-report questionnaires. NHANES has collected accelerometry data in some cycles, enabling such analyses.

Finally, examining diabetes outcomes including complications, mortality, and healthcare utilization could extend understanding beyond diagnosis to disease impact and management. Linkage with administrative health records could facilitate such research.

7.5 Conclusion

Diabetes remains a major public health challenge with significant implications for individual health outcomes and healthcare system sustainability. This research confirmed that traditional risk factors including age and obesity maintain strong associations with diabetes in contemporary data, while revealing that some hypothesized lifestyle factors showed weaker or null associations in this sample.

The persistent racial disparities observed underscore the importance of addressing health inequities through comprehensive approaches targeting both individual and structural determinants. Continued surveillance using nationally representative data like NHANES enables monitoring of trends and evaluation of intervention effectiveness.

From a methodological standpoint, this project demonstrated the value of open health data for research and education. The availability of high-quality, comprehensive health surveys enables students and researchers

worldwide to address important questions without requiring expensive primary data collection. Such resources are invaluable for developing the next generation of health data scientists.

In conclusion, this research successfully explored relationships between lifestyle factors and diabetes using current national data, identified important patterns and disparities, and provided a foundation for future work. The combination of statistical rigor, appropriate visualizations, and thoughtful interpretation demonstrates the power of data science to generate actionable insights for public health.

ANNEXURES

Annexure-I: Summary of Research Project - II

Registration No: 322201210

Name of student: Pargat Singh

Title of the Research Project - II:

Examining Lifestyle Factors and Metabolic Health Markers Associated with Diabetes Using NHANES Data (2021-2023)

Objectives:

1. Download, clean, and prepare NHANES 2021-2023 data using R programming.
2. Conduct statistical tests examining associations between lifestyle factors and diabetes.
3. Create visualizations showing metabolic marker distributions.
4. Analyze racial disparities in diabetes and metabolic health.

Results and Findings: 963 participants (18.7% diabetes). Age, BMI, glucose, HbA1c highly significant ($p < 0.001$). Sedentary behavior ($p = 0.517$) and smoking ($p = 0.884$) not significant. Lower carbohydrate intake in diabetes group. Significant HbA1c racial disparities ($p < 0.001$).

Specific outcomes: Complete R workflow implemented with documentation. Four professional visualizations created. Confirmed age and obesity as dominant risk factors. Identified persistent racial health disparities.

Learnings: Epidemiological data handling. Applied Wilcoxon, chi-square, Kruskal-Wallis tests. ggplot2 skills. Distinguished association from causation.

Challenges: Sample reduced from ~11,000 to 963. Self-reported measures limitations. Cross-sectional design constraints.

Signature of the Student: _____

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